EDADet: Encoder–Decoder Domain Augmented Alignment Detector for Tiny Objects in Remote Sensing Images

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***Abstract*— In recent years, deep learning has shown great potential in object detection applications, but it is still difficult to accurately detect tiny objects with an area proportion of less than 1% in remote sensing images. Most existing studies focus on designing complex networks to learn discriminative features of tiny objects, usually resulting in a heavy computational burden. In contrast, this article proposes an accurate and efficient single-stage detector called EDADet for tiny objects. First, domain conversion technology is used to realize cross-domain multimodal data fusion based on single-modal data input. Then, a tiny object-aware backbone is designed to extract features at different scales. Next, an encoder–decoder feature fusion (EDFF) structure is devised to achieve efficient cross-scale propagation of semantic information. Finally, a center-assist loss and an alignment self-supervised loss are adopted to alleviate the position sensitivity issue and drift of tiny objects. A series of experiments on the AI-TODv2 dataset demonstrate the effectiveness and practicality of our EDADet. It achieves state-of-the-art (SOTA) performance and surpasses the second-best method by 9.65% in AP50 and 4.86% in mAP.**

***Index Terms*— Cross-domain multimodality, encoder–decoder feature fusion (EDFF), loss function, remote sensing image, tiny object detection (TOD).**

I. Introduction

As THE foundation for various advanced visual tasks such as scene understanding, object tracking, image captioning, and visual navigation, object detection is a research hotspot in the field of computer vision [1]. Tiny object detection (TOD), as an important subfield, has great research and application value in autonomous driving, remote sensing monitoring, remote video surveillance, aerial image analysis, space monitoring, the detection of microdefects on material surfaces, and other areas.

TOD faces the following challenges.

1) The target image size is extremely small (usually smaller than 16×16 pixels), with low resolution and little structural and texture information [2].
2) Targets have a low signal-to-noise ratio, making them easily affected by factors such as illumination changes, object scale variations, and complex background environments.
3) Single-modality data provide limited feature descriptions for weak targets and are not sufficient for TOD.

After a long period of development, object detection has obtained significant progress and breakthroughs [1], [3], [4]. Nevertheless, the aforementioned characteristics of tiny objects make it highly difficult for the network to capture discriminative features, resulting in a high rate of missed detections and false detections. Consequently, TOD remains one of the most challenging subtasks in object detection.

To achieve higher performance for the detection of tiny targets, the network needs to be customized, enhanced, and designed. One of the simplest approaches is to develop a high-quality tiny object dataset [5], [6], [7], [8], [9]. Numerous experiments have shown that without changing the network architecture, a high-quality dedicated dataset can lead to better performance of the detector [10]. Dedicated datasets can avoid the interference of large-sized targets on network parameters. Meanwhile, multimodal datasets can provide multidimensional features. However, regarding the difficulty in obtaining tiny target data and the cost of annotation, the dedicated dataset solution is suboptimal.

Another approach is to enhance the discriminability and expression fusion capabilities of features. Some studies have been devoted to normalizing features of different scales. Feature pyramids [11] are proposed for multiscale feature fusion. In addition, some GAN-based methods generate high-resolution representations of tiny objects in an adversarial manner, or directly generate more tiny objects in the original image [12], [13], [14]. Though these methods improve model performance, they introduce additional computational burdens.

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Moreover, the selection of training samples has a large impact on the performance of anchor-based detectors. The position sensitivity of tiny objects makes it a great challenge to set appropriate thresholds for positive and negative sample assignments. Although some automated label assignment strategies, such as ATSS [15] and ASAS [16], have been developed, they still perform suboptimally when dealing with tiny objects.

In this article, a network architecture called EDADet is specifically designed for TOD. First, to address the issue of data and feature deficiency, a lightweight cross-domain modality transformation structure is designed to improve the discriminative features of tiny objects. It achieves pixel-level multimodal data fusion without incurring additional computational costs. Second, a feature extractor is developed specifically for tiny objects. The extractor pays more attention to the texture information of tiny objects while suppressing background interference. Then, the encoder–decoder structure integrates deep abstract information with shallow detailed information through cross-layer connections, resolving the problems of feature singularity and deficiency.

Finally, a novel loss function is designed specifically for tiny objects. In the regression branch, a center-assisted loss function is incorporated, which is position-insensitive and can continuously optimize bounding box regression, especially during the early stages of network training when predictions are unstable. In addition, dual detection heads are used in the network, and an alignment self-supervised loss function is introduced between them. This self-supervised approach constrains the deviation between two position predictions.

In summary, the contributions of this article are as follows.

1) A computationally friendly cross-domain multimodal foreground enhancement method is proposed, which does not rely on multimodal data input and directly generates cross-modal data using the domain conversion structure. Then, the generated data and the original data are combined with internal information in a pixel-level symmetric approach without introducing additional costs.
2) The feature extractor is redesigned to be aware of tiny objects. Dynamic convolution is used to capture the contextual information of tiny objects, and more computational resources are allocated to the initial stages of the network to extract detailed information of tiny objects.
3) An encoder–decoder structured feature fusion module is proposed to achieve efficient multiscale feature fusion and cross-scale propagation of contextual information.
4) The bounding box metric is analyzed, and based on this, a new loss function is designed, including center auxiliary loss and alignment self-supervision loss. The former overcomes the sensitivity of position prediction, while the latter constrains the drift in position prediction. In addition, the auxiliary detection branch can be removed after training, incurring no additional computational cost.
5) The proposed EDADet demonstrates robust performance in TOD and achieves state-of-the-art (SOTA) performance.

The rest of this article is organized as follows. Section II presents the related work on TOD and bounding box metrics. Section III introduces the overall architecture of EDADet and the proposed targeted improvement strategies in detail. Section IV conducts extensive experimental validation of the proposed improvements. Finally, Section V concludes this article.

II. Related Work

A. TOD

TOD, as a challenging and important branch in the field of object detection, has attracted substantial research attention. The main research methodologies can be divided into three major categories: feature fusion, data augmentation, and label assignment.

1) Feature Fusion: Due to their extremely small size, usually about 10 pixels and occupying less than 1/100 of the entire image area, tiny objects contain very limited discriminative features [17]. In detection tasks, shallow features and deep features are mainly used for regression tasks and classification tasks, respectively [18]. Current popular feature extractors [19], [20], [21], [22], [23], [24] generally perform multiple downsampling operations to obtain high-level semantic information suitable for the classification branch. However, due to the small size of tiny objects, boundary and texture details will be lost after multiple times of downsampling, seriously affecting the performance of the regression branch. Meanwhile, the features of tiny objects are easily contaminated by the background after convolution. To enhance the feature representation of tiny objects and improve the performance of both the classification and regression branches, a straightforward and typical approach is to fuse features of different scales and perform object detection at multiple scales. SSD [25] can deal with objects of varying sizes by combining predictions from multiple feature maps of different resolutions. FPN [11] realizes multiscale feature fusion via a top–down architecture with lateral connections and constructs fused feature maps at different scales. FSSD [26] addresses the issue of difficult multiscale feature fusion in SSD by integrating a lightweight feature fusion module that concatenates features from different layers and scales. AugFPN [27] analyzes the design defects of the feature pyramid in FPN and overcomes these defects by introducing consistency supervision [28], residual feature enhancement [21], and soft RoI selection [29].

2) Data Augmentation: Data augmentation is widely used in training vision models to minimize distributional discrepancies between training and testing datasets [30]. The upper performance limit of detectors mainly depends on the quality of sample instances in the dataset. Due to the low-quality features of tiny objects, more high-quality instance samples are needed for network training [31]. Therefore, data augmentation provides another straightforward and effective method for improving the detection performance of small samples. Data augmentation can be further refined into methods such

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as input data preprocessing [32], image feature enhancement [33], and specialized datasets [34], [35].

a) Input data preprocessing: In the field of object detection, techniques such as random flipping, random erasing [36], random cropping [37], and random sampling [38] are most commonly used [31], [39]. Mosaic [38] greatly increases the number of tiny objects by randomly scaling and stitching four images from the dataset. Hao and Zhili [40] optimized the mosaic method by analyzing the synthetic image regions, dividing them into irregular grids, and randomly filling them with a certain number of images, leading to better image synthesis capability. Meanwhile, directly sampling tiny objects and randomly copying and pasting them into different locations within an image can also enhance the image [41]. Based on this approach, AdaResampling [42] substantially reduces logical inconsistency caused by random paste operations by addressing background errors and scale differences in the sample pasting process. STDA [43] is a data augmentation framework developed based on shape deformation guidance and environmental adaptation, and it enhances the dataset by transforming real instances from the same dataset into different shapes.

b) Image feature enhancement: Generative adversarial networks (GANs) support network training by generating realistic data samples [44]. Therefore, GANs can be adopted to learn hidden augmentation spaces to assist in network decision-making. The GAN-based model in [14] learns to create high-resolution features under the guidance of a newly designed super-resolution feature discriminator. SOD-MTGAN [12] generates clear, high-resolution facial images directly from blurry, small images by guiding the generator network to restore fine details. DS-GAN [13] generates realistic tiny objects from larger ones by combining a GAN-based object generator with object segmentation, image inpainting, and image blending techniques.

c) Specialized datasets: The COCO dataset [34], a benchmark in the field of object detection, contains more than 1.5 million instance objects, and it can guide the training and validation of various advanced networks. Nevertheless, tiny objects constitute only about 40% of all instances in COCO. Mixed training using instances of varying scales can lead to feature interference and confusion, making it difficult to specifically optimize the detector for tiny objects. To address the issues of data sparsity and the lack of tiny objects, several datasets specifically designed for small object detection have been proposed, such as SOD [45], Tinyperson [46], UAVDT [47], and VisDrone [48]. However, due to their smaller scale or lack of focus on tiny objects, these datasets do not meet the requirements for training supervised TOD algorithms based on convolutional neural networks (CNNs).

3) Label Assignment: Tiny objects are highly sensitive to positional changes [5]. During the early stages of training, the values of network parameters are unstable, and anchor-based detectors generally produce anchor boxes that deviate significantly from ground-truth boxes, leading to no overlap between them. In this case, it is difficult to use the intersection over union (IoU) [49] to determine their positional relationship or guide the bounding box regression process [6]. Meanwhile,

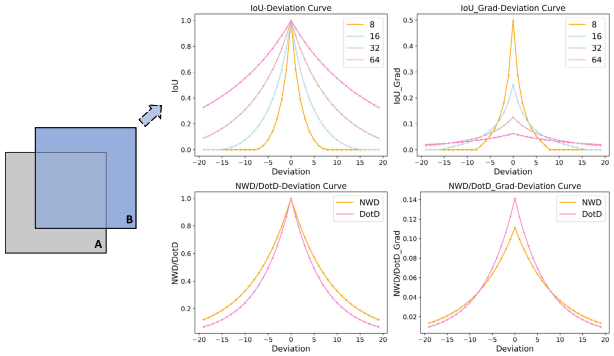


Fig. 1. Bounding box similarities and their gradients under different metrics.

detectors typically set an IoU threshold to distinguish between positive and negative samples. The sensitivity of IoU makes it challenging to set a reasonable threshold, resulting in a lack of positive samples for tiny objects [7]. To resolve the problem of positive sample deficiency for tiny objects and mitigate the instability in sample division during the regression process, many high-quality and innovative label assignment strategies have been developed.

S3FD [8] incorporates a scale compensation strategy for anchor box matching to increase the recall rate of tiny objects. The research in [50] proposed an expected maximum overlap (EMO) score to theoretically address the issue of low overlap between the bounding boxes. LTM [51] uses a maximum likelihood estimate (MLE) framework to realize anchor-free matching, breaking the limitations of IoU. ATS S [52] significantly enhances the performance of anchor-based detectors by automatically selecting positive and negative samples according to their statistical characteristics. OTA [53] views the label assignment process from a global perspective, treats it as an optimal transport (OT) problem, and achieves optimal label assignment by minimizing transportation costs.

B. Bounding Box Metrics

In object detection, bounding box metrics are used to evaluate the discrepancy between the predicted and ground-truth bounding boxes. IoU is a classic metric for object detection benchmarks, but it has limitations in assessing the positional relationship between two bounding boxes because it only works when the boxes overlap. Specifically, Fig. 1 presents the curves of several classic bounding box similarity measures and their gradients regarding positional deviations. The same sizes are maintained for boxes A and B, and B offsets from A along the diagonal direction. The objects are of four typical sizes (8, 16, 32, and 64). It can be seen that smaller boxes are more sensitive to positional deviations.

When there is no overlap between two bounding boxes, the IoU is 0, which neither reflects the positional relationship between the boxes nor generates gradients to guide bounding box optimization (as shown in the first row of Fig. 1). To address these defects, several new metrics based on IoU have been proposed. GIoU [54] introduces the concept of the minimum encircling rectangle based on IoU, and it considers not only the overlapping area but also the relative position and size of two bounding boxes. CIoU [55] combines IoU, the

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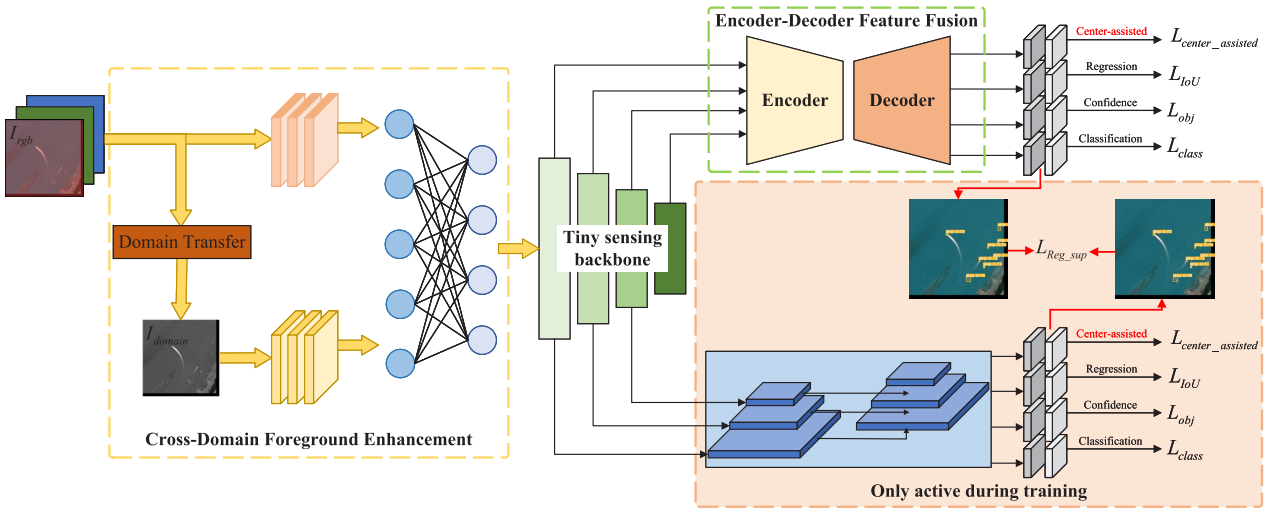


Fig. 2. Overall architecture of EDADet. The input image undergoes cross-domain enhancement, and then it is fed into a tiny object-aware backbone for feature extraction. Subsequently, the features are input into an encoder–decoder structure for fusion and center-aware loss calculation. In addition, to achieve higher performance, a self-supervised branch is incorporated during the training phase to accelerate training. During the testing phase, this self-supervised branch is eliminated to enhance inference speed to match the baseline.

distance between the central points, and the consistency of the aspect ratio. EIoU [56] considers IoU, the distance between the central points, and the width–height differences of the bounding boxes. Nevertheless, these improvements are only adjustments based on IoU and do not fundamentally change the sensitivity of IoU to tiny objects.

DotD [7] is a novel and efficient metric that directly uses the normalized Euclidean distance between the center points of two bounding boxes. NWD [57] resolves the sensitivity problem by modeling the bounding boxes with Gaussian distributions and exploiting Gaussian distribution similarity instead of the IoU score. DotD and NWD are not affected by object size, so they are particularly suitable for measuring tiny objects. As shown in the second row of Fig. 1, no matter how the bounding box size changes, the similarity measurement index curves at different scales completely coincide and will not be 0. Therefore, gradients are always generated in the whole training process for bounding box regression optimization. That is, the metric is insensitive to bounding boxes.

III. Methodology

A. Overview of the Framework

Fig. 2 demonstrates the overall architecture of EDADet, our proposed encoder–decoder domain-enhanced alignment-assisted framework for TOD. Given an RGB image $I_{\text{rgb}} \in \mathbb{R}^{H\times W\times 3}$, label $T \in \mathbb{R}^{n\times 5}$ (n denotes the number of instances in each image, and each instance consists of $[\text{class}_{id}, x, y, w, h]$), our goal is to predict the location $[x, y, w, h]$ and category class_{id} of all tiny targets in the image.

EDADet is composed of four components: cross-domain foreground enhancement (CDFE) module, tiny perception feature extractor (TPFE), encoder–decoder feature fusion (EDFF) module, and auxiliary self-supervised loss. It reduces interference from complex backgrounds, improves fusion and discriminative expression capabilities of features, and enhances the accuracy of tiny object recognition. Specifically, during training, for an input RGB image I_{rgb}, CDFE first augments the cross-domain modality and transforms I_{rgb} into $I_{\text{domain}} \in \mathbb{R}^{H\times W\times 1}$ using domain transformation operations. Then, it uses a computationally efficient pixel-level fusion strategy to combine I_{rgb} and I_{domain}, to refine features and enhance foreground information. Subsequently, the feature extractor performs continuous convolution operations on the enhanced mixed features to extract features of different levels for tiny objects. Next, EDFF receives features from different stages and performs complex feature fusion. Finally, the detection loss is calculated for the extracted features of both the backbone and the EDFF, and a self-supervised bounding box loss is incorporated to form the loss function of the network.

In our work, EDADet is trained in an end-to-end approach. During training, the aligned self-supervised loss takes the predicted bounding boxes from both the backbone and the EDFF and measures the similarity between them in a self-supervised manner. In contrast, the auxiliary self-supervised loss aims to constrain the positional deviations between the two predicted bounding boxes, thereby limiting the positional drift of tiny objects. This design effectively overcomes the training difficulty for tiny objects through self-supervision.

B. Cross-Domain Foreground Enhancement

Tiny objects lack sufficient information and have weak contrast with the background. More available distinguishing information will contribute to higher detection performance. Multimodal data fusion can fundamentally improve the spatial dimensional information of the target. However, due to the complexity of the data collection and annotation process for tiny objects, most current datasets specially designed for tiny objects contain only single-modal data. To address this issue, our study innovatively proposes CDFE, which accepts single-modal RGB images and uses domain transformation techniques to obtain cross-domain modality-augmented images. Then, the two types of data are fused after the

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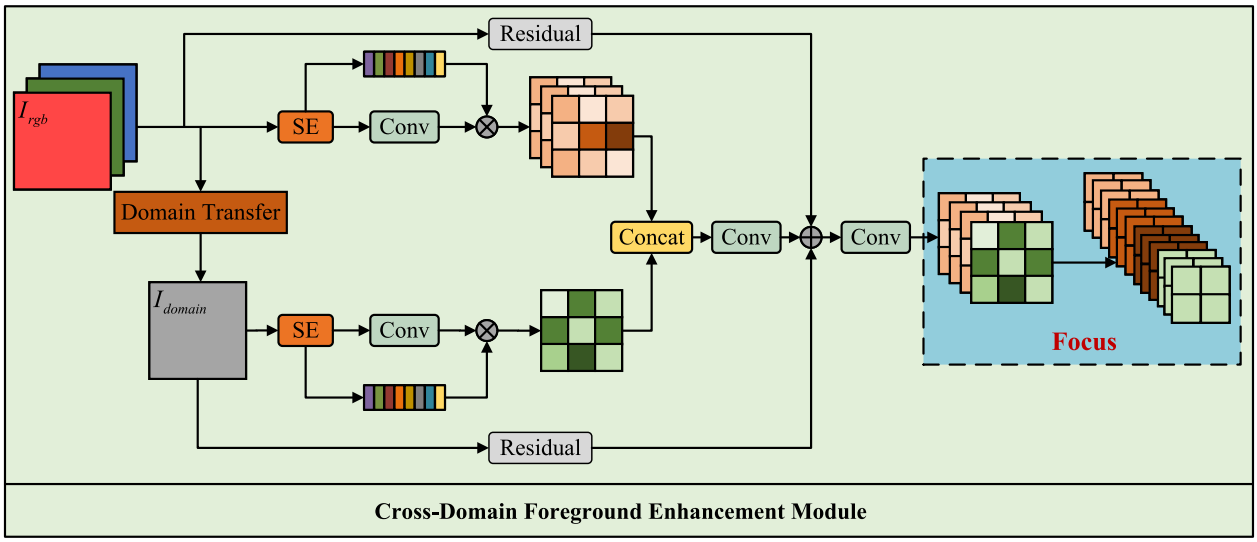


Fig. 3. Structure of the CDFE module.

extraction of foreground enhancement information under different modalities symmetrically and compactly. CDFE breaks the dependency of the model on native multimodal datasets, making it particularly suitable for vision tasks with limited data. Specifically, feature extraction and fusion are conducted at the pixel level.

As demonstrated in Fig. 3, to realize modality augmentation and pixel-level fusion, the RGB image $I_{\text{rgb}} \in \mathbb{R}^{H\times W\times 3}$ is first input into a domain transformation structure to obtain the cross-domain image $I_{\text{domain}} \in \mathbb{R}^{H\times W\times 1}$

$$I_{\text{domain}} = f_{\text{domain}}(I_{\text{rgb}}) \tag{1}$$

where f_{domain} denotes the domain conversion function. To avoid incurring additional computational costs, image grayscaling is used here for domain conversion. Then, I_{rgb} and I_{domain} are fed into two SE blocks to calculate channel attention. Since I_{domain} has only one channel, the global average pooling operation on I_{domain} is skipped, and a standard 3×3 convolution is used instead

$$F_{\text{rgb}} = f_{\text{MLP}}(f_{\text{Avgpool}}(I_{\text{rgb}})) \times I_{\text{rgb}} \tag{2}$$

$$F_{\text{domain}} = f_{3\times 3}(I_{\text{domain}}) \times I_{\text{domain}} \tag{3}$$

where f_{Avgpool} represents the global average pooling operation, f_{MLP} denotes a two-layer fully connected network, and $f_{3\times 3}$ stands for a 3×3 convolution. Then, F_{rgb} and F_{domain} are concatenated along the channel dimension to obtain the attention map for foreground importance across different domain spaces

$$F_{\text{cat}} = f_{3\times 3}(f_{\text{Concat}}(F_{\text{rgb}}, F_{\text{domain}})) \tag{4}$$

where f_{Concat} represents the concatenation along the channel dimension, and the newly obtained F_{cat} has a channel dimension of 4 and a shape of $\mathbb{R}^{H\times W\times 4}$. To integrate the foreground enhancement attention features and spatial texture detail features, the attention map is incorporated into the original input image and then fed into a 1×1 convolution

$$F_{\text{full}} = f_{1\times 1}(f_{\text{Concat}}(I_{\text{rgb}}, I_{\text{domain}}) + F_{\text{cat}}) \tag{5}$$

where $f_{1\times 1}$ refers to a 1×1 convolution. Finally, the focus structure slices F_{full} along the spatial dimension and concatenates them along the channel dimension. The reconstructed feature map has four times the number of channels of the original feature map, while its spatial dimensions are halved. This process realizes lossless downsampling while increasing the number of instances in the sample.

C. Tiny Perception Feature Extractor

The feature extractor serves as the backbone of the neural network, and it is responsible for extracting features at different levels. Increasing the network depth allows for obtaining richer and more complex semantic features while increasing the network width allows for obtaining higher fine-grained features. However, both the approaches will increase the network's complexity and computational load. To achieve optimal network performance, it is necessary to balance complexity and performance. EfficientNet [19] uses neural architecture search (NAS) technology to search for network architectures that strike an optimal balance between complexity and performance. Nevertheless, as a general feature extractor, EfficientNet is not specifically designed for tiny objects. To address this issue, this article improves EfficientNet to develop a specialized feature extractor for tiny objects.

1) Dy-MBConv: Standard 2-D convolution uses static convolution kernels to process all input data, so it lacks flexibility in handling different samples. In contrast, dynamic convolution uses an attention mechanism to calculate the weights of multiple independent convolution kernels, and it can make corresponding adjustments for different input feature maps by linearly combining different convolution kernels. The generalized dynamic convolution formula is given by

$$X_{\text{out}} = (\alpha_1 w_1 + \cdots + \alpha_n w_n) * X_{\text{in}} \tag{6}$$

where $X_{\text{in}} \in \mathbb{R}^{C_{\text{in}}\times H\times W}$ denotes the input feature, $X_{\text{out}} \in \mathbb{R}^{C_{\text{out}}\times H\times W}$ denotes the output feature, α_i stands for the attention weight, and w_i is the convolution kernel.

Inspired by the theory of dynamic convolution, this article proposes a multidimensional attention dynamic convolution

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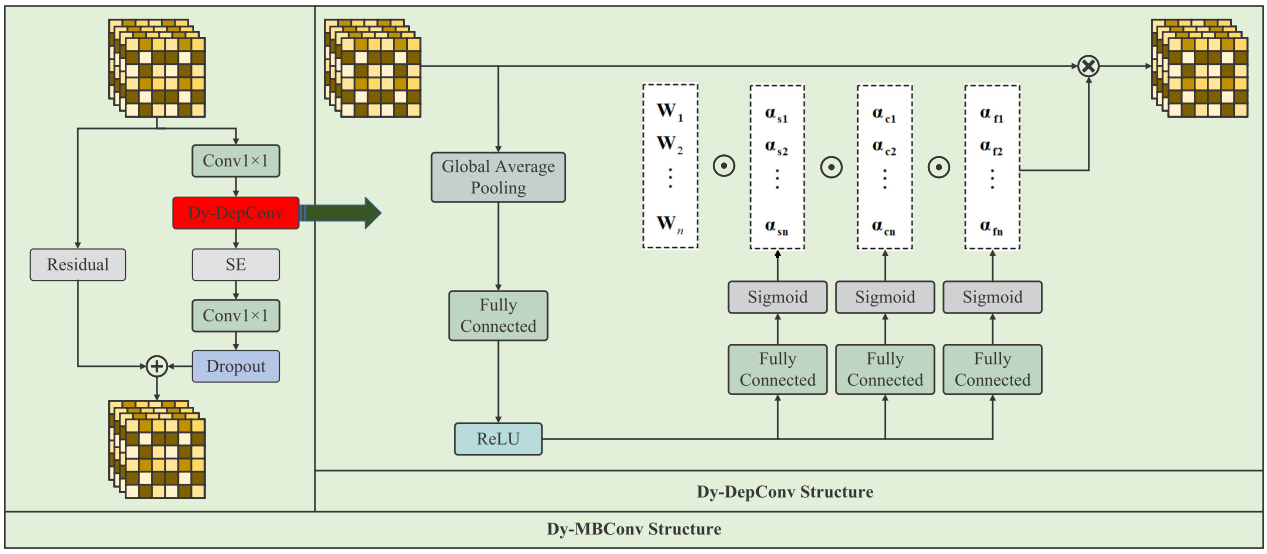


Fig. 4. Sstructure of Dy-MBConv. Dy-MBConv is the basic module of the tiny perception backbone, and its core component is Dy-DepConv.

called Dy-DepConv. Dy-DepConv calculates the attention weights of convolution kernels in different dimensions in parallel. Though it enhances the weak structural features of tiny objects and suppresses background interference, it incurs relatively low additional computational complexity. Dy-DepConv is represented as follows:

$$X_{\text{out}} = (\alpha_{f1} \cdot \alpha_{c1} \cdot \alpha_{s1} \cdot w_1 + \cdots + \alpha_{fn} \cdot \alpha_{cn} \cdot \alpha_{sn} \cdot w_n) * X_{\text{in}} \tag{7}$$

where $\alpha_{si} \in \mathbb{R}^{k\times k}$, $\alpha_{ci} \in \mathbb{R}^{c_{\text{in}}}$, and $\alpha_{fi} \in \mathbb{R}^{c_{\text{out}}}$ denote the attention introduced along the spatial dimension, input channel dimension, and output channel dimension of the convolution kernel w_i, respectively.

Fig. 4 shows the architecture of Dy-DepConv. The input feature map $X_{\text{in}} \in \mathbb{R}^{C\times W\times H}$ is compressed in spatial dimensions using a global average pooling layer and then fed into the fully connected layer and ReLU activation function

$$F_{\text{mid}} = f_{\text{ReLU}}(f_{\text{MLP}}(f_{\text{Avgpool}}(X_{\text{in}}))) \tag{8}$$

where $F_{\text{mid}} \in \mathbb{R}^{C\times 1\times 1}$. Then, three parallel branches use multilayer perceptron (MLP) of different sizes to compute attention along different dimensions. Finally, the attention weights are normalized using the Sigmoid function, and α_{si}, α_{ci}, and α_{fi} are obtained. This process can be expressed mathematically

$$\alpha_{si} = \text{Sigmoid}(f_{\text{MLP}}(F_{\text{mid}})) \tag{9}$$

$$\alpha_{ci} = \text{Sigmoid}(f_{\text{MLP}}(F_{\text{mid}})) \tag{10}$$

$$\alpha_{fi} = \text{Sigmoid}(f_{\text{MLP}}(F_{\text{mid}})). \tag{11}$$

Three different types of attention complement each other in terms of location, channel, and filter, contributing to high performance in capturing rich contextual clues for tiny objects. By replacing the standard convolution in the MBConv component of EfficientNet with Dy-DepConv, the Dy-MBConv module is obtained, as illustrated in Fig. 4. Some detailed information about the MBConv component can be found in EfficientNet [19].

TABLE I
CONFIGURATION OF THE TPFE AND EFFICIENTNET

Stage	Operator	Out Cannels	Kernel Size	Expansion Factor	Layers	Stride
1	Conv	64	3×3	-	2	2
2	MBConv	32	3×3	1	2	1
3	MBConv	48	3×3	4	3	2
4	MBConv	80	5×5	6	3	2
5	MBConv	160	3×3	4	4	2
6	MBConv	224	5×5	6	4	1
7	MBConv	384	5×5	6	5	2
8	MBConv	640	3×3	6	2	1
1	CDFE	32	-	-	-	2
	Dy-MBConv	64	3×3	1	2	1
2	Dy-MBConv	64	3×3	4	3	2
3	Dy-MBConv	128	5×5	6	4	2
4	Dy-MBConv	192	3×3	4	4	2
	Dy-MBConv	256	5×5	6	3	1
5	Dy-MBConv	384	5×5	6	1	2
	Dy-MBConv	512	3×3	6	1	1

2) Detail Enhancement: For tiny objects, detailed information is more important than semantic information. In the backbone, shallow feature maps with large dimensions contain abundant details owing to their high-resolution characteristics, so they are essential for recognizing and locating tiny objects. EfficientNet uses lightweight convolutional structures in the initial stages of the network, rapidly reduces the object size, and allocates most computational resources to the later stages of the network. This design facilitates the fast and efficient extraction of deep semantic information. However, this structure sacrifices a large amount of detailed information, which is detrimental to the detection of tiny objects.

To make the network pay more attention to tiny objects, this article modifies the macroscopic structure of EfficientNet to improve detailed information, and the resulting architecture is called EfficientNet-Tiny. Specifically, more computational resources are allocated to the high-resolution early stages of the network, and unnecessary final layers are removed. This novel design enables the extraction and retention of more detailed information for tiny objects. Table I presents the

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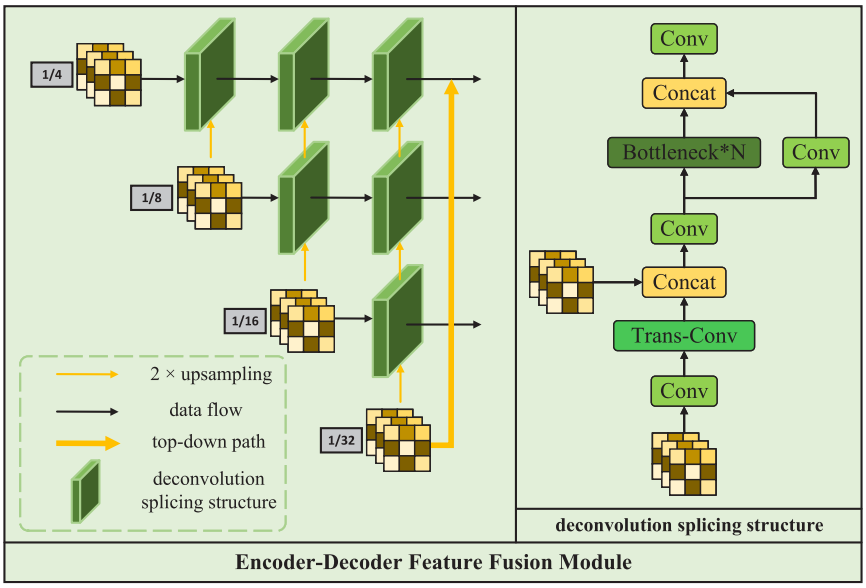


Fig. 5. Structure of EDFF module.

detailed configurations of the original EfficientNet and our EfficientNet-Tiny, where "Stage" refers to different stages of the network, "Expansion Factor" denotes the channel expansion ratio during feature map processing within the operator, and "Layers" indicates the number of repetitions of the corresponding operator.

D. Encoder–Decoder Feature Fusion

The deepest layers of a network usually contain the most abstract feature representations, which are rare in the shallower layers [58]. To better integrate contextual information with texture information and improve the granularity and discriminative power of features, this article designs a novel encoder–decoder structure inspired by Unet [59] and deep layer aggregation (DLA) [60], as illustrated in Fig. 5. Different from the original Unet structure, our encoder–decoder design includes denser connections between low-level and high-level features. Meanwhile, all simple interpolation upsampling operations are replaced with transposed convolutions to maximize the recovery of features lost due to downsampling. These modifications also help address feature alignment issues caused by upsampling. The EDFF receives feature maps from stages 2 to 5 of the backbone and involves multiple expansion paths. The high-resolution features from the backbone are combined with the upsampling outputs, and the resulting feature maps, with downsampling rates of 2 to 4, are fed into the detection head.

The EDFF is mainly composed of deconvolution stitching structures (DSSs), as demonstrated on the right side of Fig. 5. Each DSS consists of a 3×3 convolution, a deconvolution, and a $C3$ module. A feature map of shape $c \times w \times h$ is first passed through a 3×3 convolution and then input into a deconvolution layer for feature map upsampling, generating an output of shape $c \times 2w \times 2h$. Then, this output is concatenated along the channel dimension with the feature map from the previous layer of shape $c \times 2w \times 2h$ and input into the $C3$ module for mixed feature extraction. The sequential convolutional layers and concatenation operations within the DSS make it possible to assemble more precise outputs by learning from the enriched information. High-level abstract features guide the low-level features in a top–down manner, allowing the network to propagate contextual information to higher resolution layers and achieve comprehensive and discriminative feature representations.

E. Center-Assisted and Self-Supervised Loss

1) Center-Assisted Loss: Due to the position-sensitive characteristic of tiny objects, IoU is not suitable for regression optimization of tiny objects. Detectors based on IoU usually exhibit obvious performance degradation when detecting tiny objects. Specifically, during the early stages of network training, predicted bounding boxes are extremely unstable and have large positional discrepancies with ground-truth boxes, generally leading to an IoU of zero between them and resulting in a failure to generate gradients for network training. To overcome this limitation, this article introduces a center-assisted loss function in the bounding box regression branch. Since the height and width of tiny objects are far less significant than the position of their center points, our center-assisted loss function considers only the center position relationship between bounding boxes. Therefore, the center-assisted loss function is defined as follows:

$$D = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2} \tag{12}$$

$$L_{\text{center_assisted}} = 1 - e^{-D/C} \tag{13}$$

where (x_A, y_A) denotes the center coordinates of box A, (x_B, y_B) denotes the center coordinates of box B, D represents the Euclidean distance between the centers of the boxes, and C stands for the average size of instances in the dataset. Here, the Euclidean distance is normalized exponentially to ensure that the value of $L_{\text{center_assisted}}$ falls within the range [0, 1]. The definition of the new regression loss function is given below

$$L_{\text{reg}} = \alpha_1 L_{\text{center_assisted}} + \alpha_2 L_{\text{IoU}} \tag{14}$$

where α_1 and α_2 are weight coefficients, and we set $\alpha_1 = \alpha_2 = 1$. The larger the deviation of the predicted box's

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center point from the ground-truth box's center point, the stronger the correction provided by the center-assisted loss function. When the IoU loss fails and cannot provide guidance, the center-assisted loss function still enables the predicted box to continuously approach the ground-truth box, providing support to the entire training process.

Moreover, to guarantee that EDFF can better extract latent effective features, this article uses a dual detection head as shown in Fig. 1. Specifically, one detection head receives feature maps from stages 2 to 4 of the feature extraction backbone, while the other identical detection head receives the final output feature maps from EDFF at downsampling rates of 1/4, 1/8, and 1/16. The new loss function is defined as follows:

$$L_1 = \lambda_{\text{reg1}} L_{\text{reg1}} + \lambda_{\text{obj1}} L_{\text{obj1}} + \lambda_{\text{class1}} L_{\text{class1}} \tag{15}$$

$$L_2 = \lambda_{\text{reg2}} L_{\text{reg2}} + \lambda_{\text{obj2}} L_{\text{obj2}} + \lambda_{\text{class2}} L_{\text{class2}} \tag{16}$$

$$L_{\text{detection}} = L_1 + L_2 \tag{17}$$

where L_{reg}, L_{obj}, and L_{class} denote regression loss, confidence loss, and classification loss, respectively. λ_{reg}, λ_{obj}, and λ_{class} are adjustment coefficients, and we set λ_{reg}, λ_{obj}, and λ_{class} exactly the same as in YOLO.

2) Aligned Self-Supervision Loss: To further constrain the divergence of predicted bounding boxes, an aligned self-supervised loss function is introduced between the two detection heads. This helps limit the consistency of bounding box predictions and ensure their stability. Meanwhile, a metric based on Gaussian Wasserstein distance [57] is used to better evaluate the similarity between the two predicted bounding boxes. Specifically, given a bounding box with center (x, y), height h, and width w, it can be modeled using a 2-D Gaussian distribution

$$f(x|\mu, \Sigma) = \frac{\exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)}{2\pi|\Sigma|^{\frac{1}{2}}}$$

$$\mu = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \Sigma = \begin{bmatrix} w^2/4 & 0 \\ 0 & h^2/4 \end{bmatrix} \tag{18}$$

where the mean vector μ represents the center point coordinates, and the covariance matrix Σ represents the square of the half side lengths. Following the OT theory, the Wasserstein distance between the predicted bounding boxes from the two detection heads can be expressed as:

$$D_W = \|\mu_1 - \mu_2\|_2^2 + \mathbf{Tr}\left(\Sigma_1 + \Sigma_2 - 2(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2})^{1/2}\right). \tag{19}$$

Since the bounding box is a horizontal cuboid, the above expression can be simplified to

$$\begin{aligned} D_W &= \|\mu_1 - \mu_2\|_2^2 + \left\|\Sigma_1^{1/2} - \Sigma_2^{1/2}\right\|_F^2 \\ &= \left\| \left(\left[x_1, y_1, \frac{w_1}{2}, \frac{h_1}{2}\right]^T, \left[x_2, y_2, \frac{w_2}{2}, \frac{h_2}{2}\right]^T \right) \right\|_2^2 \\ &= (x_1 - x_2)^2 + (y_1 - y_2)^2 + \frac{(w_1 - w_2)^2 + (h_1 - h_2)^2}{4} \end{aligned} \tag{20}$$

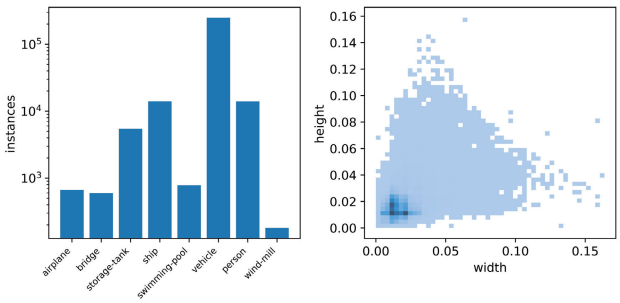


Fig. 6. Category distribution and scale distribution of instances in the AI-TODv2 dataset.

where $\|\cdot\|_F$ is the Frobenius norm. This article normalizes D_W using an exponential function to obtain the aligned self-supervised loss in the range of [0, 1]

$$L_{\text{Align_sup}} = 1 - \exp\left(-\sqrt{D_W}\right). \tag{21}$$

The aligned self-supervised loss allows for long-term continuous supervision of the bounding boxes predicted by the two detection heads, thereby enhancing the stability of the prediction. Finally, our loss function is

$$L = L_{\text{detection}} + \lambda_{\text{Align_sup}} L_{\text{Align_sup}} \tag{22}$$

where $\lambda_{\text{Align_sup}}$ is a tradeoff hyperparameter, and we set $\lambda_{\text{Align_sup}} = 0.1$.

IV. Experimental Results and Analysis

In this section, the dataset, evaluation metrics, and experimental details are first described. Then, extensive and rigorous experiments are conducted on the dataset, and our method is compared with the baseline and SOTA methods. In addition, based on the experimental results, a detailed analysis of our proposed improvements is provided.

A. Dataset

To better evaluate the detector performance on tiny objects, this article chooses the popular AI-TOD [61] and AI-TODv2 [9] aerial tiny object datasets and Tinyperson [46] dataset.

1) AI-TOD: AI-TOD [61] comes with 700 621 object instances for eight categories across 28 036 aerial images. AI-TODv2 [9] adjusted and refined the annotation files. AI-TODv2 [9] consists of 28 036 images and 752 745 instances, annotated with horizontal bounding boxes. Fig. 6 demonstrates the category distribution and the scale distribution of the instances in the dataset. The task is to detect eight common tiny objects: airplane, bridge, storage tank, ship, swimming pool, vehicle, person, and windmill. AI-TODv2 is currently the dataset with the smallest object size in aerial photography. In this dataset, each image has a resolution of 800 × 800 pixels, containing densely distributed instances with an average size of 12.7 pixels, much smaller than that in other datasets. The dataset includes complex backgrounds such as oceans, mountains, and towns, reflecting real natural scenes and posing a great challenge to the design of detection algorithms.

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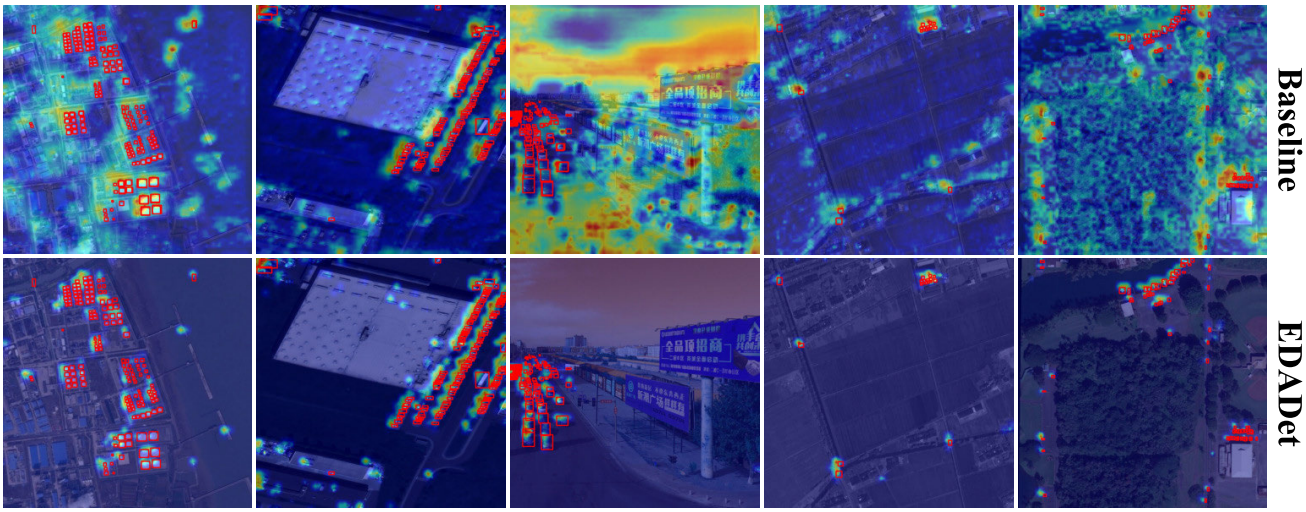


Fig. 7. Comparison of attention visualization between the baseline and our EDADet.

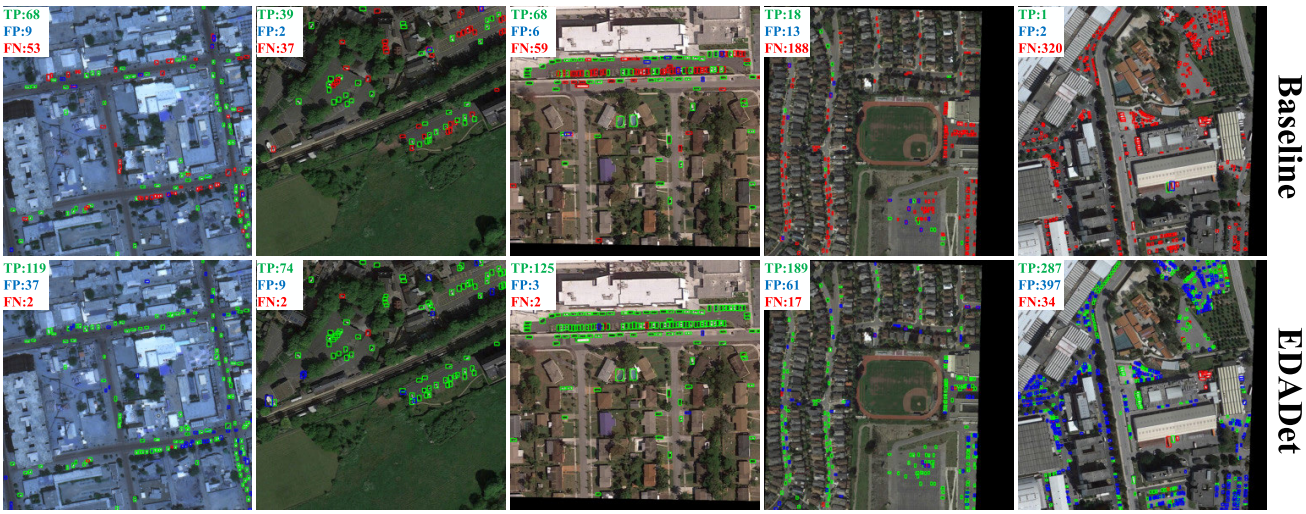


Fig. 8. Comparison of the visual detection results of the baseline and our EDADet on the AI-TOD-v2 dataset. The green, blue, and red boxes represent true-positive (TP), false-positive (FP), and false-negative (FN) predictions, respectively.

2) TinyPerson: There are 1610 labeled images and 759 non-labeled images and totally 72 651 annotations in TinyPerson [46]. In this dataset, each image has high resolution and contains densely distributed instances with an average size of 18 pixels. We just treat "normal" box as positive box, and dense images are not used for training.

B. Evaluation Metrics

In our experiments, commonly used evaluation metrics in object detection, including precision, recall, AP50, mAP, etc., are adopted to assess the consistency and differences between the detection results and the ground-truth boxes. Precision and recall are related to false positives and false negatives, respectively, and their definitions are given below

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{23}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{24}$$

AP represents the average precision of a model across different categories, and it is obtained by calculating the area under the precision–recall curve. Overall, AP considers the performance of the model in various categories and provides a more comprehensive evaluation by analyzing precision and recall. The calculation formula of AP is as follows:

$$\text{AP} = \int_0^1 P(r)dr \tag{25}$$

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N} \text{AP}_i. \tag{26}$$

In addition, this article introduces Params and GFLOPs as evaluation criteria to comprehensively evaluate and quantify the model. Specifically, Params refers to the total number of learnable weights and biases in a neural network. GFLOPs represent the number of floating-point operations the model performs in 1 s, and it serves as a measure of the model's computational complexity and performance. The two metrics provide an intuitive assessment of the model's size and computational requirements.

C. Implementation Details

The proposed framework was implemented in PyTorch, and all the experiments were conducted on the same platform. The

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TABLE II
COMPARISON OF THE SIZES AND PERFORMANCES OF DIFFERENT FEATURE EXTRACTION BACKBONES

Backbone	AP50 ↑	mAP ↑	Params ↓	GFLOPs ↓
Darknet-53	45.06	15.63	7.03	15.8
Mobilenet_small	42.74	14.87	**4.48**	**6.7**
Mobilenet_large	48.47	17.22	7.23	10.1
ConvNeXt	40.34	13	9.32	16.9
Efficientnet_b1 (stage3-5)	51.18	19.13	11.46	16.8
Efficientnet_b1 (stage2-4)	**55.03**	**21.92**	9.81	18.1
Efficientnet_v2s	54.75	20.43	20.21	48

TABLE III
PERFORMANCE COMPARISON UNDER DIFFERENT DATA AUGMENTATION STRATEGIES

Strategy	AP50 ↑	mAP ↑	Params ↓	GFLOPs ↓
-	48.09	19.5	9.81	18.1
Flipup	46.21	15.89	9.81	18.1
Fliplr	43.57	15.02	9.81	18.1
Flipup+Fliplr	40.82	13.73	9.81	18.1
Mosaic	**56.64**	**22.89**	9.81	18.1
Mosaic+Fliplr	55.03	21.92	9.81	18.1
Mosaic+Flipup+Fliplr	53.97	19.86	9.81	18.1

TABLE IV
PERFORMANCE TEST OF THE EFFICIENTNET-TINY BACKBONE

Detail Enhancement	Dy-MBConv	AP50 ↑	mAP ↑	Params ↓	GFLOPs ↓
		56.64	22.89	**9.81**	**18.1**
✓		59.59(+2.95)	25.24(+2.35)	11.26	39
✓	✓	**60.53**(+0.94)	**25.52**(+0.28)	13.07	41.5

TABLE V
PERFORMANCE TEST OF THE CDFE MODULE

Method	AP50 ↑	mAP ↑	Params ↓	GFLOPs ↓
w/o CDFE	60.53	25.52	**13.07**	**41.5**
w CDFE	**60.91**(+0.38)	**26.17**(+0.65)	13.09	43.6

TABLE VI
PERFORMANCE TEST OF THE EDFF MODULE

Method	AP50 ↑	mAP ↑	Params ↓	GFLOPs ↓
w/o EDFF	60.91	26.17	**13.09**	**43.6**
w EDFF	**64.24**(+3.33)	**28.09**(+1.92)	14.73	78.2

platform's hardware consists of four NVIDIA 4090 GPUs. The model was trained for 300 epochs with a batch size of 32. The optimizer used was stochastic gradient descent (SGD) with a momentum of 0.937 and a weight decay of $1e-5$. During the first 300 iterations, learning rate warm-up was used to linearly increase the learning rate from 0 to 0.01, followed by a cosine annealing schedule to decay it to 0.0001. Unless otherwise specified, all our experiments were conducted under the above configuration.

D. Ablation Study

The effectiveness and reliability of our proposed method were validated by designing a series of ablation experiments on the dataset.

1) Backbone Selection: To select a suitable backbone network for TOD, this article chooses feature extraction networks that are currently recognized for their outstanding performance. The original structures were retained as much as possible to load the pretrained weights provided officially as the initial weights. Table II shows the performance and size of different feature extraction backbones. EfficientNet_b1 (stages 2–4) demonstrated the best detection capability with an acceptable computational overhead, significantly outperforming other feature extraction backbones. Although EfficientNet_v2s showed similar performance, it had more than two times Params (20.21 versus 9.81) and GFLOPs (48 versus 18.1). In addition, with comparable computational demands, EfficientNet_b1 (stages 2–4) obtained 3.85% higher AP50 than EfficientNet_b1 (stages 3–5), indicating that lower level features are more suitable for small object detection. Therefore, EfficientNet_b1 (stages 2–4) was chosen as the baseline backbone for our model, and subsequent experiments were conducted on this basis to demonstrate the rationality of our framework.

2) Data Augmentation: The effect of different data preprocessing enhancement strategies in enriching the instances in the training set was investigated. Considering that the dataset consists of aerial top-view images, the augmentation strategies implemented to mimic real-world scenarios included left–right flipping, up–down flipping, mosaic augmentation, and their combinations. Table III presents the model performance under different data augmentation strategies (for subsequent experimental comparison, pretrained weights were not loaded here). The results suggest that indiscriminate data augmentation had a catastrophic negative impact on training. Both left–right and up–down flipping led to large performance drops. Only mosaic augmentation contributed to a substantial performance improvement, with AP50 increasing by 8.55% and mAP increasing by 3.39%. This is because mosaic significantly improves the diversity of tiny samples through scaling and cropping. Unless otherwise specified, subsequent experiments were conducted based on this setup.

3) Effect of TPFE: To further improve the network's performance on tiny objects, detail enhancement and dynamic convolution Dy-MBConv were adopted in the backbone network. The newly designed EfficientNet-Tiny pays more attention to fine-grained features, generating more precise representations for tiny objects. The results in Table IV indicate that EfficientNet-Tiny improves AP50 by 3.89% and mAP by 2.63%. The main performance improvement is attributed to the detail enhancement, demonstrating the importance of detailed information in TOD.

4) Effect of CDFE: Table V presents the favorable tradeoff between accuracy and complexity before and after the addition of the CDFE module. CDFE has a positive impact on the detection of tiny objects: With the addition of CDFE, the network shows better performance, with an increase in AP50 of 0.38% and an increase in mAP of 0.65%. Particularly, our CDFE module generates cross-domain images directly from the original images, with no reliance on multimodal

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TABLE VII
PERFORMANCE TEST OF THE LOSS FUNCTION

	Center-assisted Loss	Dual Heads	Aligned Self-supervised Loss	AP50 ↑	mAP ↑
EDADet				64.24	28.09
	✓			65.08	28.60
	✓	✓		66.35	28.67
	✓	✓	✓	**67.15**	**29.86**

TABLE VIII
COMPARATIVE EXPERIMENTS ON ALIGNED SELF-SUPERVISED LOSS FUNCTION

	CIoU	DIoU	GIoU	L_{Align_sup}	AP50 ↑	mAP ↑
Aligned Self-supervised Loss	✓				**62.80**	**26.17**
		✓			**62.93**	**25.85**
			✓		**63.93**	**26.33**
				✓	**67.15**	**29.86**

TABLE IX
COMPLETE RESULTS OF THE ABLATION STUDY

	TPFE	CDFE	EDFF	Loss	AP50 ↑	mAP ↑
EDADet					**56.64**	**22.89**
	✓				**60.53**	**25.52**
	✓	✓			**60.91**	**26.17**
	✓	✓	✓		**64.24**	**28.09**
	✓	✓	✓	✓	**67.15**	**29.86**

datasets, and this is appealing for vision tasks with limited data. Meanwhile, the slight increase in Params and GFLOPs of CDFE demonstrates the necessity of choosing pixel-level modality fusion instead of feature-level modality fusion. Overall, the CDFE module is generalizable and scalable, making it widely applicable to conventional single-modal datasets.

5) Effect of EDFF: The EDFF module is introduced into the network to enable deep abstract semantic features to provide global support to shallow appearance features. This module facilitates the seamless fusion of features at different levels through multiple bottom–up paths. As listed in Table VI, the inclusion of EDFF greatly improves network performance, with detection metrics AP50 and mAP increasing by 3.33% and 1.92%, respectively. Compared with the model without EDFF, Params increased by 1.64 M and GFLOPs increased by 34.6G. To better detect tiny objects, prioritizing accuracy is essential, and a small increase in parameters and computational cost is acceptable. Notably, the EDFF module is well-organized and can be easily introduced into other TOD architectures without extensive modifications.

6) Effect of Loss Function: To optimize the proposed EDADet network, several improvements are made to the loss function, including introducing a center-assisted loss function $L_{\text{center_assisted}}$ into the regression branch, using dual detection heads, and incorporating an aligned self-supervised loss function $L_{\text{Align_sup}}$. The new loss function has the following features: 1) it constrains the divergence of bounding boxes, leading to more accurate localization and 2) it reduces the sensitivity to the scale of tiny objects and addresses the inconsistency between the loss function and evaluation metrics. As listed in Table VII, the experimental results indicate that the new loss function further improves AP50 and mAP by 2.91% and 1.77%, respectively. It is noteworthy that the auxiliary detection branch (highlighted in orange in Fig. 2) can be eliminated during the inference stage, thus bringing no additional parameters or computational cost.

TABLE X
DETECTION PERFORMANCE OF DIFFERENT CATEGORIES IN THE VALIDATION SET

Class	Instances	Precision ↑	Recall ↑	AP50 ↑	mAP ↑
airplane	177	87.07	72.88	77.23	33.98
bridge	152	71.66	58.55	58.25	24.54
storage-tank	2502	86.46	84.65	87.74	45.22
ship	3828	83.75	79.30	82.26	42.36
swimming-pool	112	53.22	75.00	70.27	32.88
vehicle	60851	80.46	82.30	82.97	36.47
person	3776	69.79	33.28	39.73	13.34
Wind-mill	69	51.46	49.28	38.79	10.07
all	**71467**	**72.98**	**66.91**	**67.15**	**29.86**

Moreover, to highlight the benefit of $L_{\text{Align_sup}}$ in network training, we conducted comparative experiments with commonly used IoU-based loss functions. As shown in Table VIII, the IoU-based loss functions negatively impacted network training, with each to varying degrees compromising the detector's expected performance. In contrast, only $L_{\text{Align_sup}}$ further enhanced the detector's performance.

Tables IX and X show the total ablation study results and the detection performance of EDANet across various categories, respectively. As the improved modules were progressively added, the performance of the detector steadily increased. Due to the limited number of instances, the training with windmills and swimming pools was not adequate, resulting in the lowest accuracy. Nevertheless, high detection scores were obtained for the other categories, demonstrating EDANet's robust capability for detecting tiny objects.

E. Comparison to the SOTA

EDADet was compared with other SOTA detectors specialized for TOD on the AI-TODv2 dataset. The experimental results are listed in Table XI. Among the methods taken for comparison, Cascade R-CNN [62] and DetectoRS [64], configured with NWD-RKA or ADAS-GPM, are the current leading detectors for tiny objects. Though these methods demonstrated impressive performance, our EDADet obtained significantly superior results, setting new SOTA performance on the AI-TODv2 dataset. The experimental results suggest that EDADet improved AP50 and mAP by 9.65% and 4.86%, respectively. Our framework is specifically designed for the detection of tiny objects, so it surpasses the baseline models. First, cross-domain data synthesis is used to improve training, which effectively mitigates the issue of limited data. Second, EDADet incorporates a feature extraction backbone more suitable for tiny objects and integrates an efficient feature

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TABLE XI

PERFORMANCE COMPARISON OF THE PROPOSED EDADET DETECTOR AND OTHER DETECTORS ON THE AI-TODV2 VALIDATION SET. THE LAST ROW SHOWS OUR PROPOSED EDADET, WHICH OBTAINED THE SOTA PERFORMANCE. RED, BLUE, AND GREEN INDICATE THE TOP THREE PERFORMANCES

Method	Year	Backbone	mAP ↑	AP50 ↑	AP75 ↑	AP_{vt} ↑	AP_t ↑	AP_s ↑	AP_m ↑
Anchor-based two-stage:									
Faster R-CNN [3]	2015	ResNet-101-FPN	13.1	30.7	9.2	0.0	9.7	24.6	35.5
Cascade R-CNN [62]	2018	ResNet50-FPN	14.4	32.7	10.6	0.0	9.9	28.3	39.9
TridentNet [63]	2019	ResNet-50	9.7	23.3	6.5	0.1	5.2	20.5	32.7
DetectoRS [64]	2021	ResNet-50-RPN	15.6	35.0	11.3	0.1	11.8	28.6	**43.1**
DetectoRS w/NWD [9]	2022	ResNet50-FPN	**24.5**	**56.4**	**17.4**	**7.6**	**24.3**	**29.9**	41.4
Faster R-CNN w/NWD-RKA [9]	2022	HRNet-w32	20.5	51.5	12.4	5.8	20.3	25.4	35.7
Faster R-CNN w/RFLA [57]	2022	ResNet-50-FPN	20.9	51.8	11.8	6.0	20.9	26.0	35.9
Cascade R-CNN w/ADAS-GPM [65]	2023	ResNet-50-FPN	24.2	54.2	17.0	6.0	24.0	29.3	40.0
Faster R-CNN w/ADAS-GPM [65]	2023	ResNet-50-FPN	22.3	53.7	13.5	**7.1**	21.9	27.5	35.1
DetectoRS w/ADAS-GPM [65]	2023	ResNet-50-FPN	**25.0**	**57.5**	**17.3**	**7.1**	**24.8**	**30.8**	**41.7**
Anchor-free:									
RepPoints [66]	2019	ResNet-50-FPN	10.6	27.8	5.6	2.0	10.1	16.0	21.8
Grid R-CNN [67]	2019	ResNet-50-FPN	14.7	31.7	11.4	0.0	11.5	27.4	38.0
FOCS [1]	2019	ResNet-50-FPN	13.3	32.5	8.2	1.7	11.0	20.7	32.8
AutoAssign [15]	2020	ResNet-50	14.3	37.5	7.5	3.9	14.1	19.4	23.2
TOOD [68]	2021	ResNet-50	18.6	43.0	12.7	3.2	16.8	26.2	38.1
Anchor-based one-stage:									
SSD-512 [25]	2016	VGG-16	8.3	25.7	2.8	1.7	6.3	14.8	26.9
RetinaNet [4]	2017	ResNet-50-FPN	7.4	21.1	3.5	2.5	6.5	13.1	22.9
YOLOv3 [69]	2018	DarkNet-53	4.1	14.6	0.9	1.1	4.8	7.7	8.0
ATSS [52]	2020	ResNet-50-FPN	15.5	36.5	9.6	1.9	12.7	24.6	36.2
YOLOv5 [70]	2020	DarkNet-53	16.4	45.6	8.5	6.0	15.6	19.6	29.6
SM+ [71]	2021	ResNet-50-FPN	19.8	46.1	12.7	5.0	19.9	26.3	37.1
SSPNet [72]	2022	ResNet50	13.1	30.3	8.8	0.0	9.7	27.1	37.6
Ours:									
EDADet(ours)	2024	Efficientnet-Tiny(ours)	**29.86**	**67.15**	**24.2**	**12.3**	**30.1**	**36.1**	**42.0**

fusion module based on an encoder–decoder structure. Finally, a novel loss function is introduced into EDADet to accelerate network training and improve prediction accuracy. EDADet achieved the highest detection accuracy across objects of various scales, with significant improvements in the very tiny and tiny categories. This demonstrates the advanced nature of our algorithm.

Table XII presents the performance comparison between EDANet and other popular detectors for different categories of objects. Except for slightly lower detection performance for ships, EDANet achieved SOTA performance in all other categories. The results in Tables XI and XII illustrate the potential of our framework compared with other deep learning methods. In addition, EDANet is a single-stage detector that can perform end-to-end training and prediction. It exceeds most of the two-stage detectors in performance while maintaining lower complexity. Overall, EDANet provides a straightforward and effective solution for TOD with promising application prospects.

To demonstrate the versatility of EDADet for TOD, we conduct experiments on two additional tiny object datasets, AI-TOD and Tiny-Person, and compare them with advanced detectors on the corresponding datasets to further verify the credibility and excellent performance of our method. As shown in Tables XIII and XIV, EDADet achieved SOTA on both AI-TOD and Tiny-Person, fully demonstrating the algorithm's powerful ability to detect tiny objects.

F. Visualization

In Fig. 7, the output features of the baseline network and our EDADet are visualized to provide a more intuitive and interpretable description. To reflect the real detection performance of the network, we choose to visualize the attention output of the past few layers, that is, the last layer weight parameter before the detection head. Since the backbone inputs features of different scales into three detection heads to detect objects of different scales. Therefore, we select three attention weights at diverse scales and stretch them to the same size as the input image. The features are overlaid on the original input image and presented as attention heatmaps to demonstrate the network's focus on different regions of the image. It can be seen that the baseline network distributes its attention across the entire image but fails to capture the target objects effectively. In contrast, EDADet pays most attention to the objects to be detected, effectively suppressing the interference of complex backgrounds and obtaining clearer object features.

Fig. 8 presents the partial detection results of the baseline detector and EDADet. Compared with the baseline, EDADet significantly improves detection capability, with a substantial reduction in missed detections. This indicates that our framework, after being trained extensively on the dataset, can accurately capture discriminative features of tiny objects. At the same time, due to its excellent performance, EDADet can detect other tiny objects that are difficult to label.

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TABLE XII

PERFORMANCE COMPARISON OF THE PROPOSED EDADET DETECTOR AND OTHER POPULAR DETECTORS ON DIFFERENT CATEGORIES OF OBJECTS IN THE AI-TODV2 VALIDATION SET. THE LAST ROW SHOWS OUR PROPOSED EDADET, WHICH OBTAINED SOTA PERFORMANCE IN ALMOST ALL CATEGORIES, FAR OUTPERFORMING OTHER DETECTORS. THE LAST COLUMN SHOWS THE COMPREHENSIVE PERFORMANCE OF ALL CATEGORIES. RED, BLUE, AND GREEN INDICATE THE TOP THREE PERFORMANCE RESULTS

Method	Backbone	AI	BR	ST	SH	SP	VE	PE	WM	AP0.5↑
Anchor-based two-stage:										
Faster R-CNN [3]	ResNet-101-FPN	25.3	8.5	19.4	19.9	12.5	14.6	4.5	0.0	30.7
Cascade R-CNN [62]	ResNet50-FPN	13.8	5.5	22.6	24.5	28.2	17.4	3.4	0.0	32.7
TridentNet [63]	ResNet-50	12.2	0.0	17.9	13.5	20.0	11.9	1.9	0.0	23.3
DetectoRS [64]	ResNet-50-RPN	18.1	7.3	24.5	25.3	28.6	17.2	4.0	0.0	35.0
DetectoRS w/NWD [9]	ResNet50-FPN	24.2	17.4	40.5	42.0	29.5	28.6	8.3	5.8	56.4
Faster R-CNN w/NWD-RKA [9]	HRNet-w32	17.8	11.0	38.1	34.7	24.2	27.7	7.1	3.7	51.5
Cascade R-CNN w/ADAS-GPM [65]	ResNet-50-FPN	20.2	19.0	41.1	40.0	30.6	29.4	7.9	4.8	54.2
Faster R-CNN w/ADAS-GPM [65]	ResNet-50-FPN	19.0	15.6	38.6	37.0	27.8	27.9	7.0	5.6	53.7
DetectoRS w/ADAS-GPM [65]	ResNet-50-FPN	25.9	18.4	40.9	42.5	29.6	28.8	8.2	5.8	57.5
Anchor-free:										
RepPoints [66]	ResNet-50-FPN	0.0	0.6	26.0	25.4	9.9	19.6	3.1	0.0	27.8
Grid R-CNN [67]	ResNet-50-FPN	13.3	10.8	22.7	25.0	26.1	16.2	3.2	0.1	31.7
FOCS [1]	ResNet-50-FPN	7.9	7.4	26.3	24.1	18.8	18.9	3.3	0.0	32.5
AutoAssign [15]	ResNet-50	11.8	10.2	25.3	24.2	17.5	18.7	5.1	1.3	37.5
TOOD [68]	ResNet-50	14.6	8.5	35.3	31.5	26.2	27.2	5.1	0.6	43.0
Anchor-based one-stage:										
SSD-512 [25]	VGG-16	7.9	4.8	13.4	13.2	12.7	11.0	2.0	1.7	25.7
RetinaNet [4]	ResNet-50-FPN	2.4	7.5	13.0	18.8	2.9	12.3	2.3	0.1	21.1
YOLOv3 [69]	DarkNet-53	0.3	0.5	8.5	9.4	0.0	12.7	1.4	0.0	14.6
ATSS [52]	ResNet-50-FPN	13.7	3.4	30.2	25.6	24.2	22.5	3.9	0.1	36.5
YOLOv5 [70]	DarkNet-53	20.8	11.5	26.5	23.4	16.8	20.4	3.6	3.4	45.6
SM+ [71]	ResNet-50-FPN	13.4	10.2	38.2	33.8	25.3	27.4	7.9	2.3	46.1
SSPNet [72]	ResNet50	15.5	3.8	22.8	20.1	23.4	15.8	3.5	0.2	30.3
EDADet (ours)	Efficientnet-Tiny (ours)	33.98	24.54	45.22	42.36	32.88	36.47	13.34	10.07	67.15

TABLE XIII

PERFORMANCE COMPARISON ON AI-TOD VALIDATION SET. RED, BLUE, AND GREEN INDICATE THE TOP THREE PERFORMANCES

Method	mAP ↑	AP50 ↑	AP_{vt} ↑	AP_t ↑	AP_s ↑	AP_m ↑
TridentNet [63]	7.5	20.9	1.0	5.8	12.6	14.0
RetinaNet [4]	8.7	22.3	2.4	8.9	12.2	16.0
ATSS [52]	12.8	30.6	1.9	11.6	19.5	29.2
AutoAssign [15]	14.2	37.8	6.4	14.9	17.4	21.8
DetectoRS w/NWD [9]	23.4	53.5	8.7	23.8	28.5	36.0
Faster R-CNN w/NWD-RKA [9]	19.5	49.2	8.3	19.6	24.5	31.9
Faster R-CNN w/RFLA [57]	21.1	51.6	9.5	21.2	26.1	31.5
Cascade R-CNN w/ADAS-GPM [65]	22.1	51.3	7.6	22.1	27.4	34.4
DetectoRS w/ADAS-GPM [65]	24.1	55.2	8.9	24.0	29.3	37.2
Ours:						
EDADet(ours)	27.3	61.0	15.4	30.3	32.3	35.6

TABLE XIV

PERFORMANCE COMPARISON ON THE TINY-PERSON DATASET. RED, BLUE, AND GREEN INDICATE THE TOP THREE PERFORMANCES

Method	AP_{tiny} ↑	AP_{tiny1} ↑	AP_{tiny2} ↑	AP_{tiny3} ↑	AP_{small} ↑
Adaptive Reppoints [66]	30.5	15.0	30.3	44.3	51.0
RetinaNet [4]	33.4	11.4	38.6	46.5	48.4
Cascade R-CNN [62]	47.0	30.9	50.8	57.8	62.2
Grid R-CNN [67]	47.1	30.7	52.2	57.2	62.5
Faster R-CNN [3]	47.6	30.2	51.8	58.5	62.7
SDS-FPN	47.8	30.5	52.0	58.7	62.9
Faster R-CNN w/RFLA [57]	50.1	32.8	55.6	60.6	65.3
Ours:					
EDADet(ours)	53.4	32.7	56.5	63.5	66.4

V. CONCLUSION

This article proposes a novel detection framework called EDADet specifically for tiny objects. First, the CDFE module achieves cross-domain multimodal foreground data enhancement. Then, the backbone network processes the enhanced foreground input and extracts features of tiny objects at different scales. Next, the EDFF structure is used to achieve cross-level feature fusion and full-link propagation of contextual information. Finally, and importantly, a novel loss function is introduced, which implements position constraints and accelerates regression for tiny objects in a self-supervised manner. The experimental results indicate that EDADet significantly improves the performance of TOD and achieves an AP50 of 67.15% and an mAP of 29.86% on the AI-TODv2 dataset with lower computational cost, setting a new SOTA performance.

Our framework provides a novel TOD solution for remote sensing tasks and explores lightweight multimodal object detection under data deficiency. In the future, we will further investigate low-parameter approaches to make the framework applicable to edge platforms with limited computational resources.

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